PORTFOLIO · FABRICATION & MECHANICAL

JOHN NORTHRUP

Architectural designer who draws it, builds it, and automates it.

Permit drawing sets for city and county approval, fabrication from concept to install, and Rhino, Grasshopper, and Python tools that speed up design work.





JCNORTHRUP@GMAIL.COM JCNORTHRUP.COM

FABRICATION & MECHANICAL

01	ELF-O-MATIC-3000	FA22, MONSTER CITY STUDIOS	03
02	DETAILS, DETAILS	F25, DETAILS, DETAILS	05



01

ELF-O-MATIC-3000

FA22, MONSTER CITY STUDIOS
ROLE · TECHNICAL DESIGNER & PROJECT MANAGER

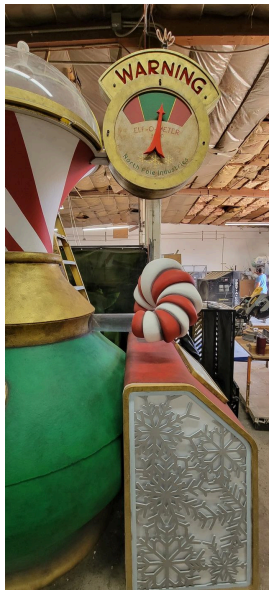
A holiday show piece built at Monster City Studios: a steam-powered candy machine with a neon marquee, a lit viewing window of tumbling balls, a STEAM grille, a pressure gauge and a candy-striped dome. Behind the theming is a working show, with lighting, moving parts and an operator panel to reset it between guests.



BUILT WORK | THE MACHINE RUNNING IN THE SHOP (VIDEO: JCNORTHRUP.COM/PROJECTS/ELF-...



BUILT WORK | MARQUEE, GAUGES AND DOME



DETAIL | WARNING GAUGE, B...



DETAIL | OPERATOR PANEL: S...



DETAIL | SERVICE HATCH WITH LABELED CONTROL SWITCHES



02

DETAILS, DETAILS

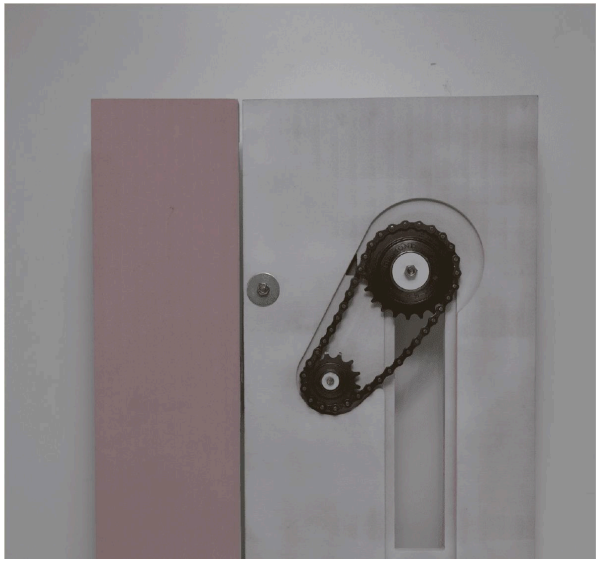
F25, DETAILS, DETAILS
ADVISOR · DWAYNE OYLER

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.

*All work made with partners Chia-Chen Li and Baizhen Yu



FOAM PROTOTYPING PROCESS



PROTOTYPE INTERIOR MECHANISM



FINAL DOOR MECHANISM, CLOSE-UP